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$$f(x) = \frac{1}{\sqrt{3+2x}} = (3+2x)^{-\frac{1}{2}}$$

$$\frac{df}{dx}(x) = -\frac{1}{2} (3+2x)^{-\frac{3}{2}} \cdot 2$$

$$\frac{d^2f}{dx^2}(x) = -1 \cdot -\frac{3}{2} (3+2x)^{-\frac{5}{2}} \cdot 2$$

$$\frac{d^3f}{dx^3}(x) = -1 \cdot -3 \cdot -\frac{5}{2} (3+2x)^{-\frac{7}{2}} \cdot 2 = \frac{-15}{\sqrt{(3+2x)^7}}$$

References